

Analyzing density dependent growth in Salmonids

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Abstract

Density-dependent growth is a phenomenon in nature among populations that can be explained using mathematical modeling. Density-dependent growth is important in population dynamics, as larger fish are favored in interactions among fish in the same population.

Accounting for external environmental factors like temperature, time of year, and habitat geography that also influence growth makes analyzing density-dependent growth among Salmonids a unique challenge. Density-dependent growth can also change based on species, even among Salmonids, as species like Atlantic Salmon are more aggressive, and thus more likely to exhibit density-dependent growth. An analysis of density-dependent growth among 3 populations of Salmonids reveals that density-dependent growth is present in these populations and varies among the three populations that were studied. This can help conservationists that stock fish populations for game and to revitalize decreased wild populations to have a better understanding of the optimal number of fish to stock, to establish healthy, sustainable populations.

Introduction

Density-dependent growth in fish is a core concept to understanding the structure and dynamics of a population. When the number of individuals in a specific area is increased, the resources in the area for an individual decrease, and growth would be slowed (Bystrom et al.

1999). Density dependence not only happens in juvenile populations, as was thought for some time, but also in fish later in life (Andersen et al. 2018). Growth in fish is important because larger fish are favored in interactions among fish in the same populations, meaning that they will have easier access to food and habitat than smaller fish competing for the same resources. (Post et al. 1999). The challenge with measuring density dependent growth is all of the external environmental factors that can also influence growth in fish, like temperature, time of year, or habitat geography.

Density dependence can vary between methods used to assess growth. When testing density dependent growth in a laboratory setting, researchers usually use higher densities of fish than what is found in nature (Matte et al. 2020). This makes it problematic to apply growth data from a lab experiment to a wild setting in a stream or a lake. Density dependent growth is also expected to be higher in laboratory experiments because survival is increased (Matte et al. 2020). Density dependence can be interesting to look at among different species. In a paper specifically studying density dependent growth in Salmonids, the more aggressive species, Atlantic Salmon, exhibited higher density dependent survival than more flexible species like Brook Trout, which exhibited higher density dependent growth than survival (Matte et al. 2020). The less aggressive species like the Brook Trout tolerate the other individuals in their population more than the aggressive species like Atlantic salmon, thus the reason for increased density dependent growth vs survival.

Density dependence is especially important among conservationists that stock fish, because it can help biologists know the optimal number of fish to stock in a certain habitat. Biologists can use this information to try to not overstock or understock with the goal of establishing a sustainable population. This analysis will focus on the impact of fish density on

the change in length among Salmonids in 3 different populations in a study done by Matte et al. (2020).

Methods

The data analyzed in this paper come from a data set in a study comprising 75 papers with individual data sets. This way, the cumulative data set will have a large sample size, in order to decrease uncertainty that comes with having too small a sample size. When choosing a data set, this study chose data sets that included either growth or survival or individual Salmonid species, among at least 2 densities (Matte et al 2020).

I analyzed this cumulative density dataset in R statistical software. To extract the response variable I wanted to test against the density and population variable, change in length, I subtracted initial length of individuals from the final length of individual. In the data, the variable “Density” has 5 unique values among the 91 observations, so during model selection, which I did using AICcmodavg from R, I treated “Density” as both a continuous and categorical variable. I made and compared models using the linear model function from R, including change in length as a function of categorical or continuous density along with population. Population is a variable with 3 different observations. In 4 of the models, I compared the additive and interactive effects of population on length change as a function of density. I assessed the residual diagnostics using base R, calculating the mean of residuals and the distribution of residuals using a Shapiro-Wilk test. I assessed the statistical significance of the best model using the “car” package (Fox J, Weisberg S 2019). I made predictions from the best model and plotted them against observed data using base R (R Core Team 2023), with an interval of confidence.

Results

The best supported model of length change was the linear model including the interactive effects of a continuous density variable and population on change. This model had a AICc score of 449.63, and an AICc weight of 0.78 (Table 1). The next best model, which incorporated the combined effects of a categorical density variable and population on change in length among Salmonids, had an AICc score of 453.28, and an AICc weight of 0.13, with a ΔAICc of 3.64. Because this was greater than the threshold of 2 used, we concluded that there was clear support for the continuous density and population model as the best model.

There were significant relationships between our response variable, change in length, and our explanatory variables, density, population, and density combined with population (Table 2). For each of these comparisons, the p-value is less than 0.05, so we reject the null hypothesis that there is no relationship between our response and explanatory variables.

In Table 3, the summary table, I can conclude the direction of the relationships between the response variable, change in length, and the explanatory variables Bob's cove population, the Freshwater population, and density is significantly negative. The p-values for the other variables; population Watern, population Freshwater combined with density, and population Watern combined with density were greater than 0.05, which means I cannot conclude that they are significant.

In each population, as density increased, change in length, or growth, decreased (Figure 1). Growth of Salmonids peaks when density is at its lowest. Each population's line has a different slope, or their growth has different density dependence. The Watern population shows the most density dependent growth, while the Freshwater population shows the least density

dependent growth. The population at Bob's cove shows the greatest growth regardless of density variation compared to the other two populations.

Discussion

I can conclude from this analysis that significant density-dependent growth is present in the salmonids used in this study. This density dependent growth varies among the three populations. As density of fish per given area, in this study meters squared, increases, there are more individuals in a population competing with each other for resources. With an increase in density, each individual will consume less food, as it is a limited resource, and possibly made to use less habitat. As a result of this, when density is high, growth will in turn be decreased among the entire population. On the other hand, when density is low, there is more food for each individual to consume. Depending on the density, competition is limited or nonexistent, and thus the Salmonids from low density populations would exhibit greater individual growth than the Salmonids from high density populations.

Findings from this analysis are consistent with previous research studying population dynamics relating to density dependent growth in past fish populations. In previous findings, as density gradient increased, the density-dependent growth also increased. In previous studies, density-dependent growth also varied among species of Salmonids and population, as it did in the three populations analyzed in this study (Matte *et al.* 2020). In the Watern population, density-dependent growth was not observed. This could be caused by an excess of food in the habitat present, or a case in a population where density-dependent survival is greater than density-dependent growth. In more aggressive species, like Atlantic Salmon, they are usually more territorial, and some individuals will get adequate food. In other species like Brook Trout,

more individuals in the population will share less of the food, which results in reduced density-dependent growth.

These findings of present density-dependent growth in populations of Salmonids are important for assessing appropriate conservation strategies. Stocking fish into streams and tributaries is a strategy conservationists use to restore populations, with the end goal to make a sustainable population. Stocking is common in Salmonids, as among anglers, they are a popular game fish. Understanding their growth rates as a function of density can help conservationists know the optimal number of fish to stock in an area to have a healthy sustaining population. If an area is overstocked, there can be wasted money, and a decrease in fish growth overall, while if an area is understocked, it may be a challenge for that population to sustain itself. Further research focusing on stocking fish for the optimal survival and growth rates according to species and population size can help conservationists have a greater understanding of how to stock for healthy fish populations.

References

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Figures and Figure Legend

Model Names	K	AICc	Δ AICc	Model Likelihood	AICc Wt.	LL	Cum. Wt.
Dens(con) * Pop	7.00	449.63	0.00	1.00	0.78	-217.14	0.78
Dens(cat) * Pop	14.00	453.28	3.64	0.16	0.13	-209.88	0.90
Dens(con) + Pop	5.00	454.08	4.44	0.11	0.08	-221.68	0.99
Dens(cat) + Pop	8.00	457.80	8.17	0.02	0.01	-220.02	1.00
Dens(con)	3.00	480.03	30.39	0.00	0.00	-236.87	1.00
Dens(cat)	6.00	484.32	34.68	0.00	0.00	-235.66	1.00
Population	4.00	504.91	55.27	0.00	0.00	-248.22	1.00
Null	2.00	518.67	69.03	0.00	0.00	-257.26	1.00

Table 1: Table from the process of model selection among 8 models, comparing the effects of density as both categorical and continuous on change in length among fish in different populations. The model with the lowest DAICc score was considered my best model if the next lowest score was greater than 2. This would have me conclude that the model comparing the combined effects of density and population on length change is the best model.

Sum sq.	Df	F value	Pr(>F)	
(Intercept)	2082.18	1.00	281.03	0.00
Population	245.89	2.00	16.59	0.00
Density	224.49	1.00	30.30	0.00
Population:Density	66.12	2.00	4.46	0.01

Table 2: Analysis of Variance table of linear model comparing combined effects of density and population on change in length in Salmonids. The rightmost column is the p-values for each row.

Estimate	Standard Error	T-value	P-value
(Intercept) Population	11.22	0.67	16.76
Freshwater	-5.71	1.06	-5.41
Watern	-0.53	0.98	-0.54
Density	-0.96	0.17	-5.50
Population Freshwater Density	0.48	0.28	1.73
Population Watern Density	-0.39	0.26	-1.48

Table 3: Summary table of coefficients and standard errors for the linear model comparing combined effects of density and population on change in length in Salmonids. The rightmost column is the p-values for each row.

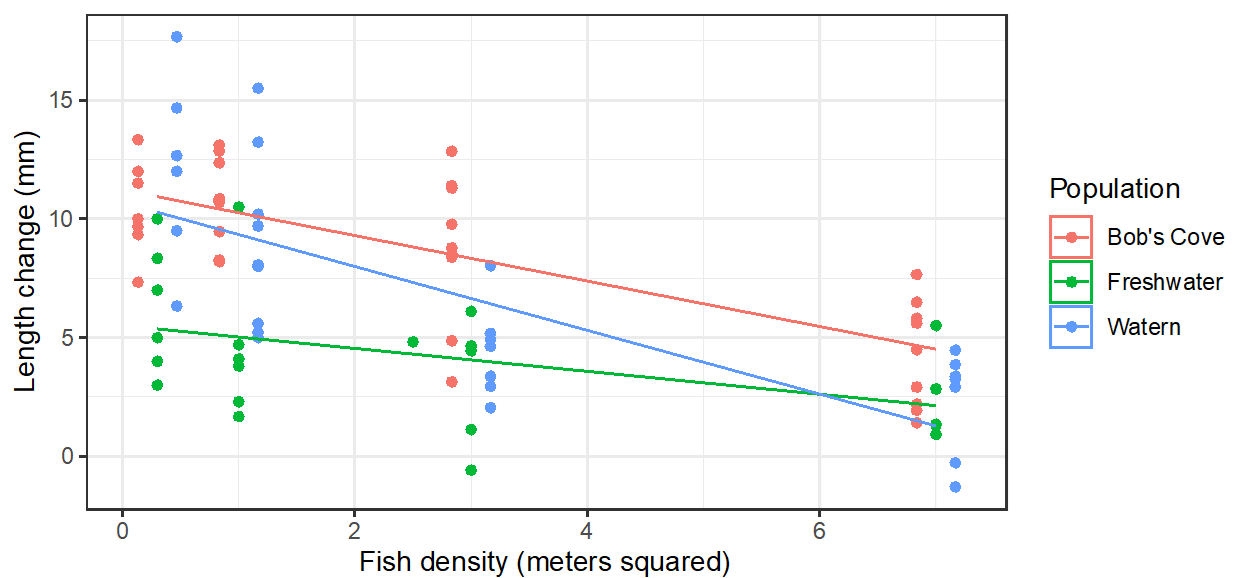


Figure 1: Change in length in millimeters in Salmonids at different densities compared among three different populations, with different colors representing the populations. Each of the different populations are at equal values where they are grouped together, a function was just used when making the graph to separate them, so every data point would be visible.